
2019-Ph-H1-Q24

Section: Electricity

Topic: Monitoring and Measuring a.c.

Question:

Which meter shows the same reading for both d.c. and the r.m.s. value of a.c.?

Answer:

Digital voltmeter gives true r.m.s. reading for a.c. and same reading for d.c.

Correct Option: E (Digital voltmeter)

Guidance for Students:

Analogue meters may not measure true r.m.s. for a.c.

Revision Tips:

- r.m.s. = equivalent d.c. voltage
 - True r.m.s. meter gives correct energy-based reading
 - r.m.s. $\neq$ peak voltage
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